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QUESTION 2 • UNIT 2 – NEURAL NETWORKS

What is the Biological Neuron? Write short note on Multilayer Feed-Forward Network with figure.

📝 5 Marks

Biological Neuron & Multilayer Feed-Forward Network

Complete answer with diagrams

Part A — What is a Biological Neuron?

A biological neuron is the fundamental unit of the nervous system in living organisms. It is responsible for transmitting information through electrical and chemical signals. The human brain contains approximately 86 billion neurons, forming an incredibly complex network.

Three Main Parts of a Biological Neuron

1. Soma (Cell Body)

Contains the **nucleus** and is responsible for maintaining the cell's functions. It processes incoming signals and determines whether to fire an output signal.

2. Dendrites

Branch-like structures that receive signals from other neurons and transmit them to the soma. A neuron may have thousands of dendrites.

3. Axon

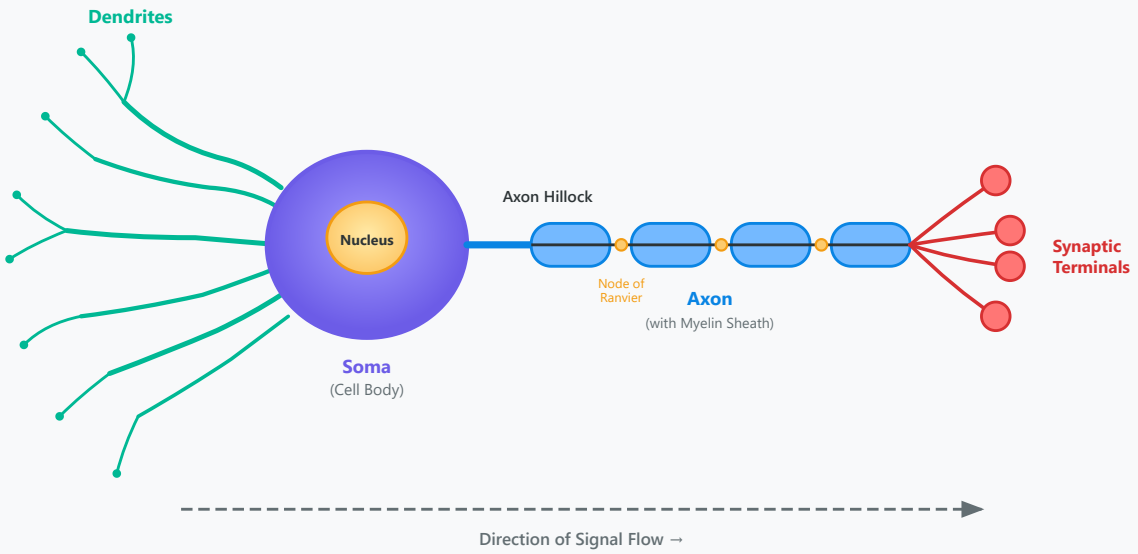
A **long, thin structure** that carries electrical impulses away from the soma to other neurons, muscles, or glands. Ends in small bulb-like structures called *synapses*.

4. Synapse

The **junction** between two neurons where signal transmission occurs via *neurotransmitters* (chemical messengers). It

connects the axon terminal of one neuron to the dendrite of another.

Figure 1 — Structure of a Biological Neuron



A biological neuron: signals arrive at dendrites, are processed in the soma (cell body), and travel along the axon to synaptic terminals that connect to other neurons.

Key Features of Biological Neurons

- Neurons communicate through **synapses** using chemical messengers called *neurotransmitters*.
- Electrical impulses are generated when the input signal exceeds a certain **threshold**.
- Neurons can be **excitatory** (stimulating other neurons) or **inhibitory** (preventing other neurons from firing).
- The biological neuron is the **inspiration** for the artificial neuron (perceptron) in deep learning.

Biological Neuron vs. Artificial Neuron

Biological Neuron	Artificial Neuron
Dendrites receive input signals	Input layer receives input features
Synaptic strengths modulate signals	Weights modulate input values
Soma processes & sums signals	Summation function (Σ) computes weighted sum
Threshold determines firing	Activation function determines output
Axon transmits output signal	Output layer produces result

Part B — Multilayer Feed-Forward Network

A **Multilayer Feed-Forward Network** (MLFF) is a type of artificial neural network where data flows in **one direction only** — from the input layer to the output layer — through one or more hidden layers. It is widely used in deep learning for tasks such as classification, regression, and pattern recognition.

Structure of the Network

Input Layer

Receives input data (features) and passes it to the next layer. The number of neurons equals the number of input features.

Hidden Layer(s)

Perform intermediate computations and transformations. Neurons apply **weights** and **activation functions** to learn complex patterns.

Output Layer

Produces the final output (predictions) based on computations performed in the

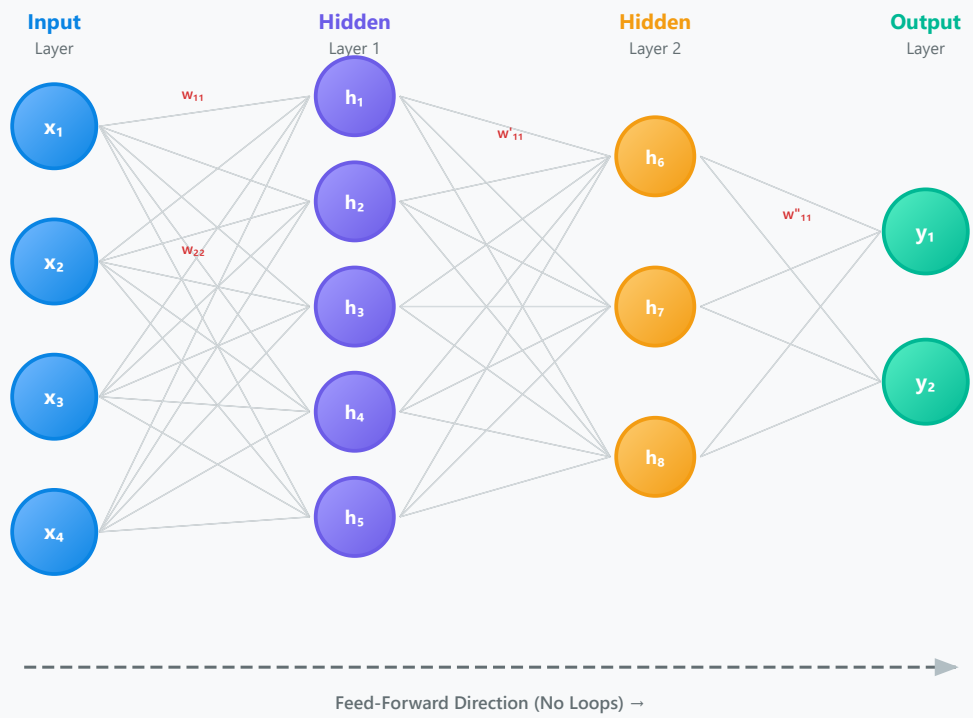
Weighted Links

Every neuron in one layer is connected to every neuron in the next layer via **weighted**

previous layers.

connections adjusted during training.

Figure 2 — Multilayer Feed-Forward Neural Network



A multilayer feed-forward network with 4 inputs, two hidden layers (5 and 3 neurons), and 2 outputs. Data flows strictly left-to-right with no feedback loops.

Key Features of Multilayer Feed-Forward Networks

- **Feed-Forward:** Data flows in *one direction only*, from input to output, without loops or cycles.
- **Hidden Layers:** Allow the network to learn *complex patterns* and *non-linear relationships* that a single-layer perceptron cannot.
- **Activation Functions:** Non-linear functions (e.g., ReLU, Sigmoid, Tanh) are applied at each neuron to enable learning of complex mappings.

- **Fully Connected:** Every neuron in one layer is connected to every neuron in the next layer (dense connections).
- **Training:** Weights are adjusted during training using *backpropagation* and gradient descent to minimize the error between predicted and actual outputs.

Applications

- **Image Recognition** — classifying objects in images (e.g., MNIST digit recognition)
- **Natural Language Processing** — sentiment analysis, text classification
- **Predictive Modeling** — stock price forecasting, weather prediction
- **Speech Recognition** — converting spoken language to text

Summary

Aspect	Biological Neuron	Multilayer Feed-Forward Network
Unit	Single nerve cell	Multiple artificial neurons in layers
Input	Dendrites receive signals	Input layer receives feature vectors
Processing	Soma sums & thresholds	Hidden layers apply weights + activation
Output	Axon transmits signal	Output layer produces predictions
Learning	Synaptic plasticity	Backpropagation & gradient descent

Aspect	Biological Neuron	Multilayer Feed-Forward Network
Direction	Dendrites → Soma → Axon	Input → Hidden → Output (feed-forward)